Algorithmic Bias

What Is Algorithmic Bias?

Algorithmic bias occurs when an AI system produces systematically unfair outcomes for certain groups — racial, gender, age, or socioeconomic. AI has no innate sense of fairness; it replicates and often amplifies the prejudices encoded in its training data.

> *Key principle: AI reflects the world as it was in the training data, not the world as it should be.*

Root Causes of Bias

Historical bias: Training data reflects past discrimination — e.g., historical hiring data underrepresents women in tech leadership.

Sampling bias: Training data doesn't represent all groups equally — facial recognition trained mostly on lighter-skinned faces underperforms on darker skin.

Label bias: Human annotators bring their own biases to labelling tasks.

Feedback loops: A biased model makes biased decisions, generating biased data, which trains an even more biased next model.

Real-World Examples

- COMPAS recidivism tool: predicted higher re-offending risk for Black defendants than white defendants with similar profiles.
- Amazon's CV-screening AI: downgraded resumes containing the word 'women's' (as in 'women's chess club') — scrapped in 2018.
- Facial recognition: NIST study found error rates up to 100x higher for dark-skinned women vs. light-skinned men.
- Healthcare: model allocating care resources used cost as a proxy for need — Black patients received worse care allocation.

Mitigation Strategies

Diverse data collection: Actively curate training data to represent all demographic groups.

Fairness metrics: Measure demographic parity, equalized odds, and calibration across protected groups.

Bias audits: Regular third-party audits of model decisions before and after deployment.

Explainability (XAI): Use interpretable models or explainability tools (SHAP, LIME) to surface why decisions were made.

Human-in-the-loop: High-stakes decisions (credit, parole, hiring) should retain human review.

Key Takeaways

- AI amplifies biases present in training data — it has no innate fairness.
- Bias sources: historical data, sampling imbalance, label bias, feedback loops.
- Mitigation: diverse data, fairness metrics, audits, explainability, human oversight.

Quick-Revision Questions

- Explain why AI systems can be biased even when designers have good intentions.
- Describe two real-world cases of algorithmic bias and their consequences.
- What is a fairness metric? Name two.